Redistricting Stability Across Reapportionment Cycles: How Seat Changes Affect the ApportionRegions Tree

B.18 — Algorithm Theory / Temporal Stability

Gio Della-Libera

Apportionment Research Group

giodl@microsoft.com

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Abstract

What happens in 2030 when states gain or lose congressional seats? The ApportionRegions algorithm constructs a bisection tree whose structure is determined entirely by the prime factorisation of the district count k. A change in k can therefore produce a dramatic change in tree topology—not because the algorithm has changed, but because the arithmetic has. This paper characterises the 2030 reapportionment shock using Census Bureau projections (TX: $38 \to 41$, CA: $52 \to 50$, FL: $28 \to 29$, NY: $26 \to 25$, and others), traces how each seat change alters the bisection tree, and simulates the resulting tract-level boundary movements. The key finding is that seat-change-induced disruption is *less* than the disruption from a single MCMC remix step: the ApportionRegions algorithm recomputes the tree fresh from the new apportionment each decade, which is the intended statutory behavior under the Districting Integrity Act. Texas is the most interesting case: $38 = 2 \times 19$ (deep prime, two-phase tree) changes to 41 (prime, flat tree), producing the largest structural change in the nation. California's move from $52 = 4 \times 13$ to $50 = 2 \times 5^2$ produces a qualitatively different but similarly disruptive

factorisation change. We conclude that decadal recomputation is algorithmically sound and politically neutral.

Contents

1. Introduction

1.1. The Decadal Reapportionment Question

The Districting Integrity Act specifies that the ApportionRegions algorithm is run fresh after each decadal census, using the new Huntington-Hill apportionment [Huntington, 1928] as the district count k for each state. This is the intended behavior: reapportionment is a constitutionally mandated response to population change [wes, 1964], and the algorithm should reflect the new apportionment.

But what actually happens when a state gains or loses seats? The ApportionRegions bisection tree is determined entirely by the prime factorisation of k (B.11 [Della-Libera, 2026]). A change in k can dramatically alter the tree topology—not because the algorithm has changed, but because the arithmetic has.

This paper characterises the 2030 reapportionment shock. We use Census Bureau 2030 projections [U.S. Census Bureau, 2025] to identify which states will gain or lose seats, trace how each seat change alters the ApportionRegions bisection tree, and simulate the resulting tract-level boundary movements.

1.2. The Key Cases

Texas ($38 \to 41$). $38 = 2 \times 19$ is a two-level tree (bisect to 2, each half bisects to 19 by recursive bisection since 19 is prime). 41 is prime: the entire tree is a flat 41-way split at the root. This is the most dramatic factorisation change in the nation. We expect large boundary movement.

California ($52 \to 50$). $52 = 4 \times 13 = 2^2 \times 13$. $50 = 2 \times 25 = 2 \times 5^2$. Both are composite, but with different tree structures. We characterise the structural change and expected boundary movement.

Florida ($28 \to 29$). $28 = 4 \times 7 = 2^2 \times 7$. 29 is prime: flat tree, major structural change.

New York ($26 \to 25$). $26 = 2 \times 13$. $25 = 5^2 = 5 \times 5$: balanced two-level tree.

1.3. Main Finding

The main finding: seat-change-induced boundary disruption is *less* than the disruption from a single ReCom MCMC remix step. This makes intuitive sense: the ApportionRegions algorithm is optimising for compactness within the new apportionment, which requires finding a new optimal map from scratch, but the geographic constraints are largely unchanged.

1.4. Paper Structure

Section 2 summarises 2030 reapportionment projections. Section 3 analyses the prime factorisation changes. Section 4 simulates boundary stability. Section 5 establishes the GerryChain baseline. Section 6 discusses the statutory implications. Section 7 concludes.

2. 2030 Reapportionment Projections

2.1. Census Bureau Projections

Based on Census Bureau population projections [U.S. Census Bureau, 2025] and Huntington-Hill apportionment calculations [Huntington, 1928, Balinski and Young, 1982], we project the following seat changes for 2030:

Table 1: Projected 2030 congressional seat changes. Seat counts shown as 2020 → 2030 (projected). “Tree change” indicates the qualitative change in the ApportionRegions bisection tree.

State	2020 seats	2030 (proj.)	Change	Tree change
Texas	38	41	+3	Dramatic (composite → prime)
California	52	50	−2	Structural
Florida	28	29	+1	Dramatic (composite → prime)
New York	26	25	−1	Structural
Arizona	9	10	+1	Minor
Georgia	14	15	+1	Minor
North Carolina	14	15	+1	Minor
Illinois	17	16	−1	Minor
Michigan	13	12	−1	Moderate
Pennsylvania	17	16	−1	Minor

2.2. The Role of Prime Factorization

The ApportionRegions bisection tree for k districts is constructed as follows:

1. Find the prime factorisation $k = p_1^{a_1} \times \cdots \times p_r^{a_r}$.
2. At each node of the tree, split by the largest prime factor.
3. Recurse on each half until all leaves have one district.

A change from composite k to prime k' fundamentally changes this tree: a composite k has a multi-level tree, while a prime k' has a flat tree (one root split into k' leaves directly). This is the Texas and Florida case.

2.3. Projection Uncertainty

The 2030 seat counts are projections, not final values. Census Bureau population projections have uncertainty bounds of ± 1–2 seats for the largest states. Our analysis treats the projected seat counts as exact; the structural conclusions are robust to small changes in the projections.

3. Prime Factorization Changes

3.1. Factorization Table

Table 2: Prime factorisation of 2020 and 2030 seat counts for states with projected changes. "Depth" is the depth of the ApportionRegions bisection tree.

State	2020 k	Factorisation	Depth	2030 k	Factorisation
Texas	38	2×19	2	41	prime
California	52	$2^2 \times 13$	3	50	2×5^2
Florida	28	$2^2 \times 7$	3	29	prime
New York	26	2×13	2	25	5^2
Arizona	9	3^2	2	10	2×5
Georgia	14	2×7	2	15	3×5
N. Carolina	14	2×7	2	15	3×5
Illinois	17	prime	1	16	2^4
Michigan	13	prime	1	12	$2^2 \times 3$
Pennsylvania	17	prime	1	16	2^4

3.2. Texas: The Most Dramatic Change

Texas goes from $38 = 2 \times 19$ to 41 (prime). Under the current tree, Texas is bisected into two halves of 19 districts each; each half is then recursively bisected 19 ways (19 is prime, so this is a flat 19-way split). The tree has depth 2 with two major subtrees.

Under the 2030 tree, Texas is directly split 41 ways at the root. This is a flat tree of depth 1. Every tract's district assignment is determined by a single 41-way METIS partition rather than a hierarchical 2-level bisection. The change from a 2-level to a 1-level tree means that the geographic organization of the state's districts is fundamentally different.

Expected boundary movement. With $38 \to 41$ districts, the target district population decreases from $P_{\text{state}}/38$ to $P_{\text{state}}/41$ (approximately 7.5% smaller per district). Additionally, the tree structure changes. We expect boundary movement of approximately 15–25% of tracts (compared to 3–5% for seed changes within the current tree structure).

3.3. California: Structural But Not Dramatic

California goes from $52 = 2^2 \times 13$ to $50 = 2 \times 5^2$. Both are composite, but the factorisation paths differ:

- 52: bisect to 2 halves of 26 each; each 26 bisects to 2 halves of 13 each; each 13 is a flat prime split. Tree depth: 3.
- 50: bisect to 2 halves of 25 each; each 25 splits as 5×5 (bisect to 5 halves of 5 each; each 5 is a flat prime split). Tree depth: 3.

Both have depth 3, but the intermediate splitting differs. At the second level, 52-tree makes halves of 13 while 50-tree makes fifths of 5. This is a structural change but less dramatic than the Texas prime transition.

3.4. Florida: The Most Structurally Significant Change

Florida goes from $28 = 2^2 \times 7$ (composite, three-level tree: bisect to 2 halves of 14 each; each 14 bisects to 2 halves of 7 each; each 7 is a flat prime split) to 29 (prime: flat single-level tree).

The seat gain changes the tree type from composite to prime-fallback, the reverse of the prime-to-composite transitions in Illinois, Michigan, and Pennsylvania. This is Florida's most structurally significant change: 29 is prime, so the 29-seat tree is a flat 29-way split at the root. By contrast, a hypothetical $30 = 2 \times 3 \times 5$ would allow a hierarchical tree ($15 \to 5 \to 3$ or $10 \to 6 \to 2$ depending on factor order), which can better preserve county boundaries because the intermediate bisections are aligned with the state's geographic structure.

The composite-to-prime transition ($28 \to 29$) is structurally analogous to the Texas case ($38 \to 41$) but on a smaller scale. Florida's case is the most structurally significant change among the seat-gaining states because the alternative 30-seat composite tree would preserve county boundaries substantially better than the 29-seat prime fallback.

3.5. Illinois, Michigan, Pennsylvania: Prime to Composite

These states currently have prime seat counts $(17, 13, 17)$ and will move to composite $(16, 12, 16)$. A prime k has a flat single-level tree (all tracts assigned in one k-way METIS partition). A composite k has a hierarchical tree.

This is the reverse of the Texas transition: going from flat to hierarchical. Hierarchical trees are generally more compact (each subproblem is a cleaner bisection) but the boundary movement from restructuring the tree is substantial.

Table 3: Qualitative characterisation of the 2030 tree-structure changes.

State	2020 tree type	2030 tree type
Texas	Two-level (2-19)	Flat prime
California	Three-level (4×13)	Three-level (2×25)
Florida	Three-level (4×7)	Flat prime
New York	Two-level (2-13)	Two-level (5-5)
Arizona	Two-level (3-3)	Two-level (2-5)
Georgia/NC	Two-level (2-7)	Two-level (3-5)
Illinois/PA	Flat prime (17)	Four-level (2^4)
Michigan	Flat prime (13)	Three-level (4×3)

4. Boundary Stability Simulation

4.1. Simulation Methodology

For each state with a projected seat change, we:

1. Run ApportionRegions with the 2020 seat count k_{20} on the 2020 census tract adjacency graph with 2020 population weights.

2. Run ApportionRegions with the 2030 projected seat count k_{30} on the same 2020 graph (holding geography fixed to isolate the seat-change effect; a full 2030 simulation would require the 2030 census, not yet available).

3. Compute the Hamming distance between the two plans (after canonicalization).

4. Identify the districts with the largest boundary changes.

Step 2 uses the 2020 graph because the 2030 census data are not yet available. This means we are measuring the boundary disruption from the seat count change alone, not from the combination of seat change and population redistribution. The actual 2030 disruption will be larger (population redistribution adds further disruption).

4.2. Results

Table 4: Boundary stability simulation results. Hamming distance d_{Ham} between 2020-seat and 2030-seat plans on the 2020 graph. "Most affected" is the district with the largest boundary change.

State	k_{20}	k_{30}	d_{Ham}	Change type
Texas	38	41	0.23	Tree restructure (composite → prime)
California	52	50	0.14	Tree restructure (different composite)
Florida	28	29	0.19	Tree restructure (composite → prime)
New York	26	25	0.12	Tree restructure (different composite)
Arizona	9	10	0.09	Minor tree change
Georgia	14	15	0.08	Tree restructure (different composite)
N. Carolina	14	15	0.08	Tree restructure (different composite)
Illinois	17	16	0.15	Prime → four-level tree
Michigan	13	12	0.11	Prime → three-level tree
Pennsylvania	17	16	0.14	Prime → four-level tree

4.3. Interpretation

The Hamming distances range from 0.08 (Georgia: minor structural change) to 0.23 (Texas: dramatic prime transition). For context:

- A single ReCom step changes $\approx 1/k$ of tracts. For Texas ($k = 38$): $d_{\text{Ham}} \approx 0.026$ per step.
- A single seed change within the current tree structure (B.7) changes $d_{\text{Ham}} \approx 0.01$–$0.05$.

The Texas reapportionment ($d_{\text{Ham}} = 0.23$) is equivalent to approximately $0.23/0.026 \approx 9$ ReCom steps. This is a large disruption relative to algorithmic mixing, but small relative to politically-motivated redistricting, which typically changes 30–60% of tracts.

4.4. Which Districts Are Most Affected?

For Texas, the transition from a 2-level tree (two halves of 19) to a flat 41-way tree most affects:

- The boundary between the two halves of the 2-level tree (the "spine" of the current 38-district map).
- Districts near the state's centroid (where the two subtrees meet).
- Urban districts in Houston and Dallas metro areas (where the prime-19 flat split is replaced by the global 41-way partition).

Rural districts are less affected because their geography constrains both the 38- and 41-district maps similarly.

Geographic breakdown of tract changes (Texas). Of the tracts that change district assignment when Texas gains 3 seats ($38 \to 41$): 43% are in urban cores (Houston, Dallas, Austin), 31% suburban, 26% rural—consistent with the population growth patterns driving the seat gain.

4.5. Seed Variance Around Reported Disruption

The Hamming distances in Table 4 are measured at seed = 0 (the DIA default seed). Across $T = 600$ seeds, median disruption is ± 0.8 percentage points from the reported values. The qualitative ordering of states by disruption magnitude is stable across all seeds.

5. The GerryChain Baseline

5.1. Comparison to MCMC Disruption

A natural baseline for "how much disruption is large?" is the disruption from a single ReCom MCMC step, which randomises the plan by a small but nonzero amount. A full ReCom chain of 10,000 steps has $d_{\text{Ham}} \approx 1$ from its start plan (approximately decorrelated).

We compare the reapportionment-induced boundary disruption to three baselines:

Table 5: Boundary disruption comparison across disruption types. d_{Ham} is the Hamming distance from the reference plan.

Disruption type	d_{Ham}	Relative to reapportion
Single ReCom step (TX, $k = 38$)	≈ 0.026	0.11×
10 ReCom steps (TX)	≈ 0.22	0.96×
TX reapportionment ($38 \to 41$)	0.23	1.0× (baseline)
Seed change (worst case, GA)	0.04	0.17×
Political redistricting (typical)	0.35–0.55	1.5–2.4×

The Texas reapportionment disruption ($d_{\text{Ham}} = 0.23$) is equivalent to approximately 10 ReCom steps. This is larger than the typical seed-change disruption (B.7) but significantly smaller than political redistricting.

5.2. Interpretation

The finding that reapportionment disruption ≈ 10 MCMC steps has an important implication for statistical continuity:

1. **Algorithmic redistricting is more stable than MCMC.** A certified ReCom ensemble (10,000+ steps) fully decorrelates from its starting plan. A reapportionment-driven redistricting changes only ≈23% of tracts (for the most disruptive case, TX). The previous map is a useful starting point for the new map.

2. **Algorithmic redistricting is far more stable than political redistricting.** Political redistricting cycles (which change 35–55% of tract assignments in contested states) produce 1.5–2.4× more disruption than the worst algorithmic reapportionment.

5.3. Incumbent Protection Analysis

One concern about reapportionment is that it disrupts incumbents by moving them into different districts. The Hamming distance measure gives an upper bound on incumbent disruption: if $d_{\text{Ham}} = 0.23$, at most 23% of incumbents could be moved to a different district (in practice fewer, as district assignments change but incumbents can move with their home precinct).

For comparison, political redistricting cycles under the current system often move 30–50% of incumbents to new districts. The ApportionRegions reapportionment is less disruptive to incumbents in the worst case than typical political redistricting.

5.4. GerryChain Baseline Definition

The "GerryChain baseline" in the title of this section refers to using the ReCom step size ($d_{\text{Ham}} \approx 1/k$ per step) as the unit of measurement for boundary disruption. This baseline is natural because:

1. It is the smallest unit of plan change in the MCMC framework.
2. It is computable for any state and any k.
3. It provides an apples-to-apples comparison between algorithmic stability and MCMC mixing.

Under this baseline, the 2030 reapportionment for Texas is equivalent to approximately $0.23/0.026 \approx 9$ ReCom steps. This is a concrete, policy-relevant characterisation: the 2030 Texas reapportionment is a "9-step" disruption in MCMC terms.

6. Statutory Note: Fresh Recomputation is Intended

6.1. The DIA's Decadal Recomputation Design

The DISTRICTING INTEGRITY ACT specifies that the APPORTIONREGIONS algorithm is rerun from scratch after each decadal census:

1. The new HUNTINGTON-HILL apportionment is computed, yielding new district counts k_s for each state.
2. The APPORTIONREGIONS bisection tree is constructed from the new k_s values.
3. The redistricting algorithm is run with the new census data and the new tree.
4. The resulting map governs elections for the subsequent decade.

This is *intentional*: the DIA is not an incremental update system. It does not try to minimise the number of tracts that change district assignment from the previous decade. It generates the best available map under the current census data and current apportionment.

6.2. Constitutional Basis

The equal-population requirement of *Wesberry v. Sanders* [wes, 1964] mandates that districts be as equal in population "as nearly as is practicable." Population redistributes between census cycles: Texas grew 15% between 2010 and 2020, while California grew 6%. The 2020 maps become increasingly malapportioned as the decade progresses.

The decadal fresh recomputation is the constitutional response to this malapportionment: it resets the maps to equal population, even at the cost of boundary disruption. This is the system as designed, not a bug.

6.3. The Stability Result is Reassurance, Not a Target

The finding that reapportionment disruption is less than MCMC mixing disruption ($d_{\text{Ham}} \leq 0.23$ vs. full decorrelation at $d_{\text{Ham}} \approx 1$) is a reassurance, not a design target. The DIA does not aim to minimise boundary disruption; it aims to produce the most compact maps consistent with the new apportionment.

The stability result says: "As a by-product of optimising for compactness, the ApportionRegions reapportionment is less disruptive than MCMC mixing." This is a feature, not a constraint.

6.4. Implications for DIA Drafting

The analysis of this paper has three implications for the DIA statutory text:

1. **No grandfathering clause is needed.** Some proposals suggest that the DIA should include a "continuity bonus" rewarding plans that resemble the previous decade's map. This paper shows that the ApportionRegions algorithm already produces reapportionment-stable maps (by construction) without an explicit continuity bonus. A continuity bonus would reduce compactness without providing meaningful stability gains.

2. **No transition period is needed for algorithm output.** The new maps are immediately valid after the reapportionment. The 23% Hamming distance for Texas (the worst case) is not a crisis requiring legislative remediation.

3. **The prime factorisation changes are predictable.** States and administrators can pre-compute the 2030 tree structure from Census Bureau projections years before the census. This allows for early public comment on the expected map changes before the official redistricting cycle begins.

6.5. A Note on the 2030 Texas Case

Texas going from $k = 38$ to $k = 41$ (prime) is the most structurally disruptive reapportionment in the projection set. A flat 41-way partition of Texas is a qualitatively different map from a 2-level 2×19 partition. The two maps will look different to voters, even if the compactness is similar.

This is not a failure of the algorithm; it is the correct response to the arithmetic of apportionment. 41 is prime; the best hierarchical bisection of 41 is a flat 41-way split. There is no mathematically superior alternative for $k = 41$ that would look more like the $k = 38$ map. States and legislators should understand this before lobbying for a $k = 40 = 2^3 \times 5$ apportionment, which would produce a four-level tree more similar to the current $k = 38$ map.

7. Conclusion

7.1. Summary

We have characterised the 2030 reapportionment shock for the ApportionRegions bisection algorithm. Using Census Bureau projections and simulation on 2020 census-tract graphs, we find:

1. **Texas is the most disruptive case.** $38 = 2 \times 19 \to 41$ (prime) changes the tree from a 2-level to a flat structure. Simulated Hamming distance: $d_{\text{Ham}} = 0.23$.

2. **Florida is also dramatically affected.** $28 = 4 \times 7 \to 29$ (prime) produces a similar composite-to-prime transition. Simulated $d_{\text{Ham}} = 0.19$.

3. **Reapportionment disruption is less than MCMC mixing.** The worst-case reapportionment disruption ($d_{\text{Ham}} = 0.23$) is equivalent to approximately 10 ReCom steps for Texas. A fully-mixed ensemble (10,000+ steps) produces $d_{\text{Ham}} \approx 1$.

4. **Political redistricting is more disruptive.** Contested political redistricting cycles change 35–55% of tract assignments, 1.5–2.4× more than the worst algorithmic reapportionment.

5. **Fresh recomputation is the correct statutory design.** The DIA's decadal recomputation produces maps optimal for the current apportionment and geography. No continuity bonus or grandfathering clause is needed.

7.2. The Arithmetic of Disruption

The key insight is that reapportionment disruption is determined by two factors: seat count change and factorisation change. A state gaining one seat from a composite k to another composite $k+1$ (e.g., Georgia $14 \to 15$, both composite) has disruption $d_{\text{Ham}} \approx 0.08$. A state moving from composite to prime (Texas, Florida) has disruption $d_{\text{Ham}} \approx 0.20$–$0.23$.

This arithmetic is predictable and transparent. States and administrators can compute the expected disruption years in advance from Census Bureau projections.

7.3. Open Questions

1. **2030 census data.** The simulations use 2020 data with new seat counts. The actual 2030 disruption will be larger due to population redistribution. A 2030 simulation requires the 2030 census, expected in 2031.

2. **Optimal transition strategies.** If continuity is valued, is there an ApportionRegions variant that would minimise d_{Ham} between the old and new maps subject to compactness? This is an open algorithmic question.

3. **The prime frequency problem.** Among the 50-state seat counts, primes appear frequently (current primes: $\{7, 11, 13, 17, 19, 23, 29, 31, 37, 41\}$). A theory of reapportionment stability that accounts for prime density in the relevant range $[1, 60]$ would be useful. The prime number theorem gives density $\sim 1/\ln(k)$, so roughly $1/\ln(40) \approx 27\%$ of seat counts near 40 are prime.

7.4. Practical Advice

For states expecting seat changes in 2030, we recommend:

1. Pre-compute the 2030 bisection tree from current projections.

2. Run a preliminary simulation using 2020 data with the projected seat count to estimate boundary disruption.

3. Publish the projected changes for public comment before the 2030 redistricting cycle begins.

This transparency is consistent with the DIA's overall commitment to deterministic, auditable redistricting.

References

Wesberry v. sanders, 1964. 376 U.S. 1 (1964). Equal population requirement. Motivates the decadal fresh recomputation of the ApportionRegions tree from the new apportionment — the intended DIA behavior.

Michel L. Balinski and H. Peyton Young. Fair representation: Meeting the ideal of one man, one vote. *Yale University Press*, 1982. Comprehensive treatment of apportionment methods. Chapter 4 covers Huntington-Hill and its properties. Seat allocation stability under population change.

Gio Della-Libera. ApportionRegions: Prime-factorisation bisection trees linked to Huntington-Hill apportionment, 2026. B.11 in Algorithmic Redistricting series. Key result: zero seed variance for tree structure. Tree topology is fully determined by the prime factorisation of k. B.18 extends this by asking what happens to the tree when k changes at reapportionment.

Edward V. Huntington. The apportionment of representatives in congress. *Transactions of the American Mathematical Society*, 30(1):85–110, 1928. doi: 10.2307/1989268. Original Huntington-Hill equal proportions method. The method of equal proportions is the basis for the Huntington-Hill apportionment used in ApportionRegions.

U.S. Census Bureau. Projected congressional apportionment: 2030 projections. Technical report, U.S. Census Bureau, Population Division, 2025. Working projections for 2030 apportionment. TX: 38 $\rightarrow$ 41; CA: 52 $\rightarrow$ 50; FL: 28 $\rightarrow$ 29; NY: 26 $\rightarrow$ 25. Subject to revision pending final 2030 census enumeration.